



Intelligent Stock Market Recommender Systems: A Review of Predictive Models, LLMs, and Decision Frameworks

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ABSTRACT

The evolution of approaches to predicting trends in financial markets has involved several stages, starting from autoregressive statistics, going further to deep sequences, graphs, fusion of sentiment with LLMs, and finally arriving at large language model (LLM) reasoning layers. One area where there has been a lack is the gap between prediction and recommendation, however, because features such as transaction costs, diversification, and investor risk profile are not taken into consideration, a prediction model may indicate a reasonable directionality for held-out samples but be uneconomical to apply when including those factors. This paper presents a survey of the current state of the art for financial trend prediction for the 2025–2026 horizon and eight methodological families, including the classical machine learning methods, recurrent sequences, attention and transformers, graph neural networks, sentiment and LLM-based fusion, explainable AI, reinforcement-learning-based trading agents, and recommender-system-specific approaches. The review shows that predictive accuracy gains have become incremental with respect to the complexity of the architecture needed to achieve them and that the evaluation methodology is now only just catching up with what is important economically and that the number of systems for which the explicit intent is that of a recommendation is still a minority compared to those that are only forecast. In this space, we motivate a design framework for a stock recommender system where prediction is not an end goal, but is one of many inputs, and we outline the open challenges which such a system must still address before it can be trusted with real money.

I. Introduction

Machine learning researchers simply, reflexively, will grab an LSTM, a transformer, pump in a few years of OHLCV data, and pull out an RMSE. Machine learning researchers will, reflexively, grab an LSTM, grab a transformer, pump in a few years of OHLCV, and pull out an RMSE. If you ask an actual investor what they want, they probably won't say that, they'll say a short list of assets that are reasonably sized and have a reason to be there. This paper is concerned with the difference between these two answers.

Research in predicting trends has really made progress. Recurrent architectures are used instead of recurrent architectures, single-series architectures are replaced by graph representations, and in the previous year, by large language models, which act as sentiment extractors and feature generators, and, more daringly, as the reasoning mechanism behind trading decisions (Chen & Kawashima, 2026). All of these changes precipitated a rush of papers promising increased accuracy, reduced error rates or better risk-adjusted returns on some criterion. What has not been quite as developed, and, of course, not to the same degree, is the work that applies a forecasting system and looks at the results of that forecasting system as a "recommendation problem" – one that must be ranked in relation to other recommendations, sized within a portfolio, explained to some person who will have to take some action on it.

This paper has three aims. Firstly, it will provide an overview of the current state of the financial trend prediction research field, using the methodology families that are most common in the literature of 2025-2026 (as published in IEEE, Springer, and ACM). Secondly, it puts forward a model which is going to reveal the difference between prediction and recommendation, by arranging the literature mentioned above, on two different axes, one of prediction and one of recommendation, which is orthogonal to the first one. Thirdly, it illustrates a way to apply such a model to define desired characteristics of a stock recommendation system in the real world.

It is here that a statement of the reason for this discrepancy is in order, rather than merely a note to that effect. The academic advantages of a clear and tractable accuracy number from a public benchmark, as opposed to a squalid claim about the usefulness of a portfolio, which is hard to assess without swapping in infrastructure and in doing so getting the transaction costs right. It is obvious that a paper can reduce the directional accuracy of predictions on S&P 500 index by half a percent, whereas a paper claiming to improve the risk adjusted portfolio returns of some type of investor is more difficult to compare with previous work and is likely to be more skeptical by the reviewers. In our view, one of the principal reasons for the sparse research described in Section IV-G that is published for recommendations is this difference between publishable and useful. The rest of the paper is organized as follows. Section II places this review in the context of previous reviews and defines its focus. The conceptual framework is introduced in Section III. Section IV presents a literature review, in a category-by-category format. In Section V, the categories are directly compared and the implications of the comparison discussed. In Section VI, open challenges are listed. Section VII concludes.

II. Related Work and Scope

Some recent survey papers have covered related areas. Kiruthika's 2025 survey of the recurring difficulties in the deep learning approach to stock price prediction – non-stationarity, regime shifts, large amounts of noise relative to signals, and the issue of over-fitting historical prices where only prices are considered – is an outstanding guide to understanding why there continue to be incremental advances, not quantum leaps, in this field (Kiruthika, 2025). Two other survey papers cover graph-based approaches to finance [mentioned repeatedly in Section IV-D below], and applications of LLM to quantitative finance (Chen & Kawashima, Sentiment-aware stock price prediction with transformer and LLM-generated formulaic alpha, 2026). These surveys provide

greater coverage of their respective subjects than is possible here; however, neither covers the topic of recommendation, where accuracy is the bottom line. Where this survey stands out in scope in a very particular way is its deliberate sampling of eight architectural families, instead of focusing in detail on one like other surveys, to enable comparison across a number of criteria. That comparison is, to our knowledge, less commonly attempted than a within-family deep dive, and it is the comparison itself that motivates the conceptual framework in Section III.

Of course, there is also a practical point to sampling in such a broad way instead of diving deep on something specific, like graph neural networks. The recommendation system that will actually be designed will not be able to focus on just one type of architecture; it will need the prediction layer, a way to incorporate relational structures between the suggested assets, a way to handle the sentiments and/or the textual content, a way to turn the predictions into the positions, and a layer of justifying what it produces. That is five out of eight types of architectures described in this article working together, not competing, while a review that covered only one of them would hardly solve any design problem this paper is focused on. Of course, we are not saying that the breadth-first strategy used in our research covers all the relevant papers in each sub-field; the focused review of graph-based methods, for example, would cover their literature much more deeply than Section IV-D does.

III. Conceptual Framework: From Prediction to Recommendation

A forecasting system answers the question "What will this stock do?" A recommender system answers a different question: "What should this investor hold, and why?" The two are related but not the same, and conflating them is, in our reading of the literature, the single most common design flaw in systems that describe themselves as investment tools.

The literature is organized by two axes. The first of these is predictive complexity, meaning essentially how well a system models the structure of financial data. The second axis is recommendation readiness – whether the system's predictions take the form of an actionable set of investments rather than just a number or category associated with a single security. This is mapped out in Table I of Section V, using the eight families identified above.

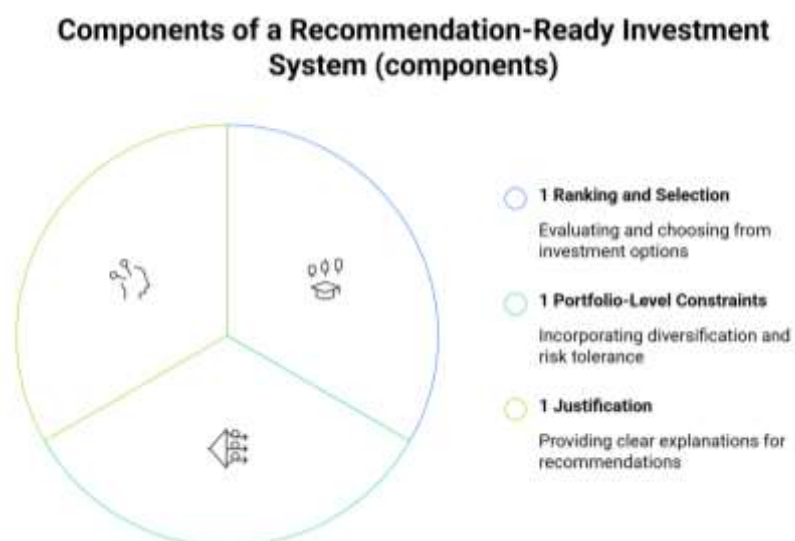


Figure 1: Components of Recommendation-ready investment system

The framework has three components that a fully recommendation-ready system needs, none of which a pure forecaster is obligated to provide:

1. Candidate ranking and selection from an entire universe of possible candidates. A forecaster has the job of forecasting one time series (or a few time series at once), whereas a recommender must rank a large number of candidates before making a selection, a completely different objective than that of minimizing point forecast errors (Guo, Sanz-Cruzado, & McCreddie, 2025).
- 2 Portfolio-level constraints. Most of the architectures discussed in Section IV were not actually tested at the portfolio level, and they are not properties of an individual forecast, but rather of the portfolio.
- 3 Justification. For an industry focused on compliance, a recommendation that is not comprehensible to the individual tasked with carrying it out lacks actionability – and thus the reason for explaining being part of the structure, and not merely diagnosis, in Section IV-F.

Nothing about this should detract from the literature on prediction, which has provided real improvements in accuracy to a recommender system. The point of the framework is more subtle: it distinguishes between the ability of a model to predict well and the ability of a system to recommend well, and the review in Section IV is designed so as to keep this distinction clear category by category.

Price predictor does not equal recommender system by definition, and a temperature gauge is not a bad thermostat. However, what the usefulness of such a framework lies in is not in comparison of the families according to one single measure.

IV. Thematic Review of Existing Systems

A. Statistical and Classical Machine-Learning Baselines

Prior to the advent of deep learning as the de facto method, trend prediction was done using autoregressive and econometrics models including ARIMA, GARCH-family volatility models, exponential smoothing, and classifier-based approaches including support vector machine, decision tree, and shallow neural network based on technical indicators. However, these approaches continue to be relevant not due to their performance but since every modern hybrid model needs to prove itself better in order to validate extra complexity. The paper by Kayit and Ismail from 2025 serves a good reference point on this topic; with a nearly ten-year history of S&P 500 market, the authors compared logistic regression, random forest, decision tree, SVM, k-NN, AdaBoost, and Lasso-LSTM hybrid models and used Kruskal-Wallis and Dunn post-hoc tests to show that performance difference between single classifier and hybrid ensemble approaches is statistically significant (Kayit & Ismail, 2025). This can be taken as an adequate overview of classical models: simple, affordable, and always outperformed in presence of volatility and non-linear signals.

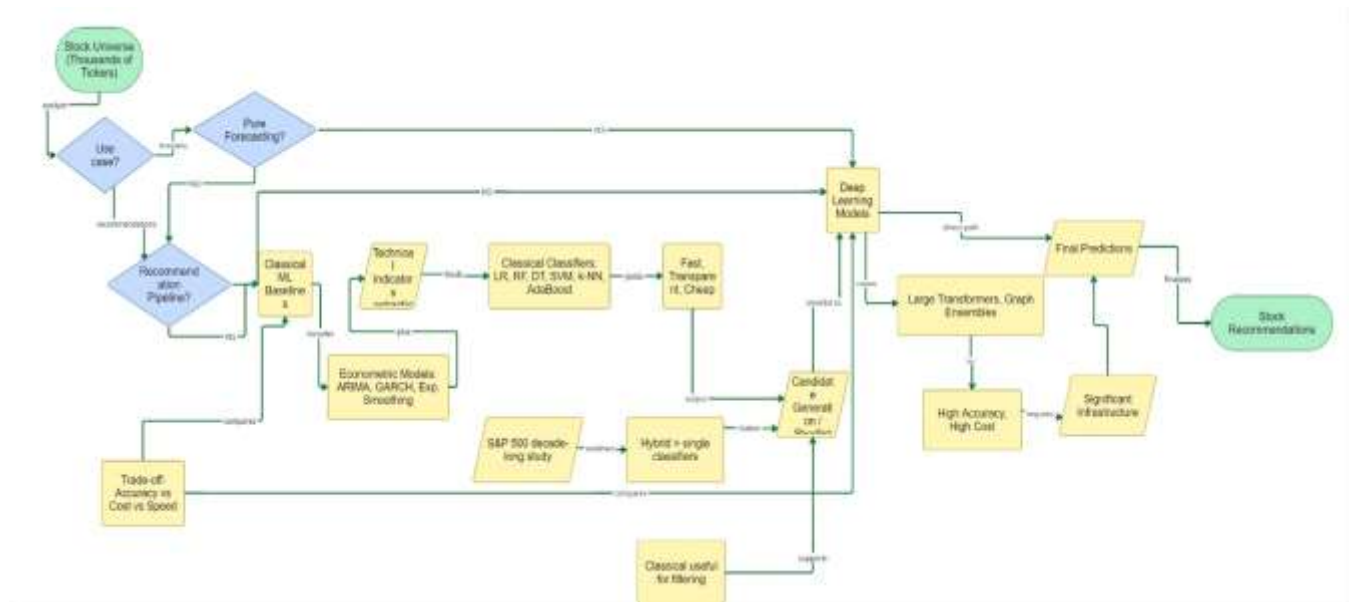


Figure 2 Overview of Classical Statistical and Machine-Learning Approaches Used as Baselines in Stock Trend Prediction

Having said that, however, it would be an error to discount all classical benchmarks completely for the particular application of a recommendation engine as compared to the forecasting one. Logistic Regression or Random Forest trained on just a few technical indicators will run fast enough to be recalculated intra-day across thousands of potential stock ticker codes, which is unlikely the case for a large Transformer model or graph-based ensemble, without a significant infrastructure spend. For the task of candidate generation – i.e., reducing the pool of several thousand stocks into a smaller list which then should be processed by a more costly model – the statistically weaker yet computation-light models presented above may still find themselves in demand.

B. Deep Sequence Models and Their Successors

Long-term dependency is still a challenge for the field of deep learning, while LSTM and GRU remain the most successful architectures to accommodate it with their gating mechanism, respectively. Still, the most recent research looks at the LSTM model not as an end-to-end solution but only as a part of a greater system. In the A-share market, China, for instance, Wang et al. (Wang, Xie, Duan, Xiong, & Chen, 2026) were able to combine their predictions into a long-short portfolio by changing the criteria from RMSE to a new set of criteria of economic performance based on a nonlinear identification factor approach (Diff-RMSE), along with designing a finance-related weighted evaluation index (WEI), and the direction of the prediction and the risk in the tails were also considered. The authors' result of better drawdown and long-short spread of LSTM and Transformer models than linear and tree models is particularly beneficial due to the similarity of the criterion used for validating the models to real-life trading systems.

What this research approach does not resolve also should be noted. The list of issues presented by Kiruthika – non-stationarity, regime change, noise and weak signal – are all problems which cannot be fully dealt with even with further improvements in recurrent depth and complexity, since they relate to the nature of the data-generation process (Kiruthika, 2025). There is an

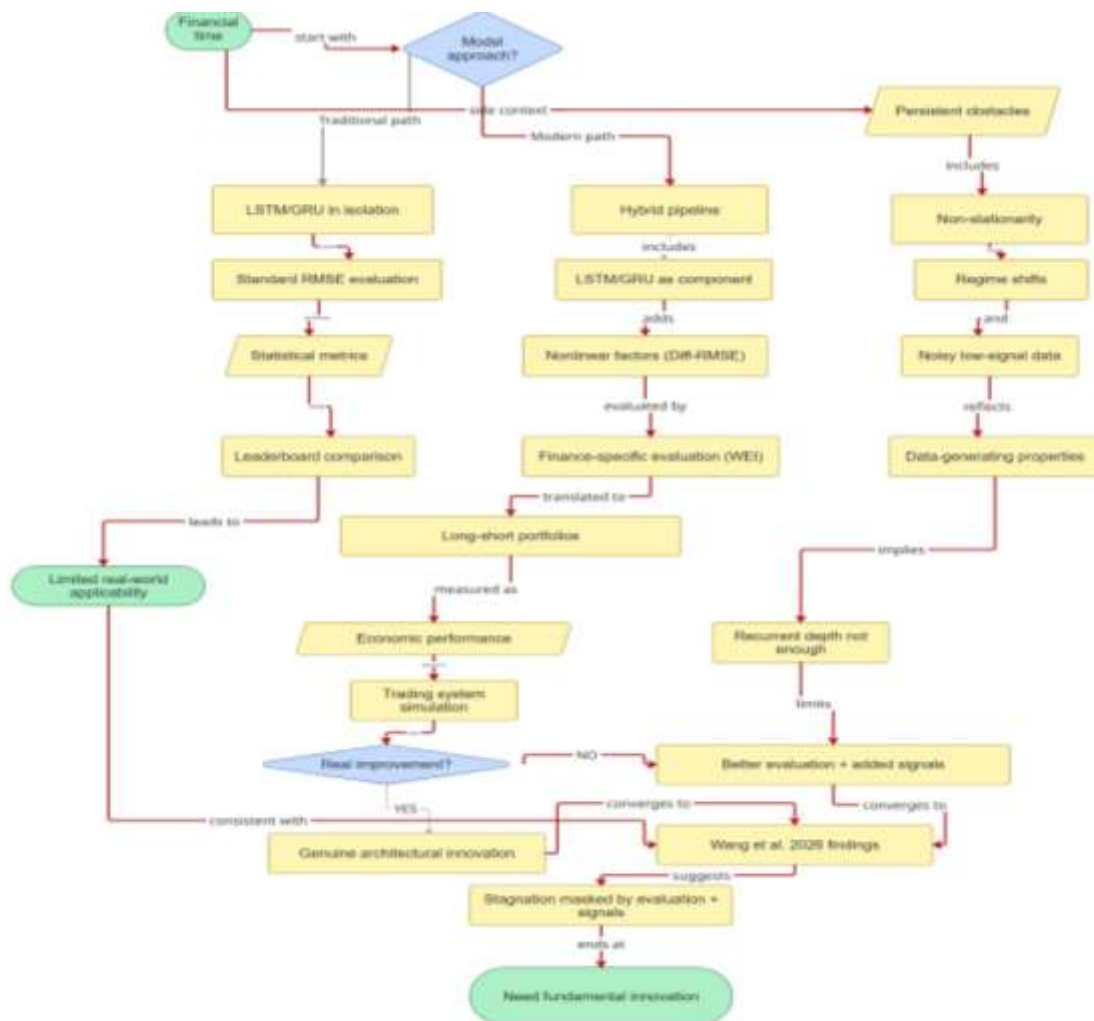


Figure 3 LSTM Architecture Stagnation Analysis

ongoing, somewhat uncomfortable trend in this sub-literature wherein innovations in the architecture of LSTM and GRU-based variants continue beyond any useful return on genuinely novel information; many of the advances that are being made using "deep sequence models" in the 2025-2026 literature are, upon further investigation, advances in how the model is being evaluated and additional signals, such as graph structure, sentiment, and LLM-derived features, on top of a recurrent architecture that hasn't really been improved at all.

C. Attention, Transformers, and LLM-Augmented Prediction

Since self-attention removes or is used together with recurrent layers, there is no need of a bottleneck. Because no need of a bottle neck, self-attention has replaced or been used with recurrent layers. When the evaluation was not limited to performance indicators, another feature of Wang et al.'s model was discovered: the transformer module had a positive but strong bias in their forecasts (Wang, Xie, Duan, Xiong, & Chen, 2026).

New variations on the transformer are now beginning to embed a large language model in the mix. In a study by Nejatbakhsh and Aliasgari, LSTM networks and transformer-based attention, combined

with sentiment analysis of news using FinBERT, and a privacy-preserving federated learning layer, were compared against several conventional forecasting methods, such as ARIMA, SARIMAX, Prophet, random forest, and support vector regression, and the hybrid consistently outperformed the conventional methods in capturing temporal dependencies across a basket of technology stocks (Nejabatbakhsh & Aliasgari, 2025). The addition of federated learning, in and of itself, is informative — it's a sign that the constraints of sharing data across institutions are no longer considered a constraint, but part of the design, in systems meant for deployment, not just benchmarking.

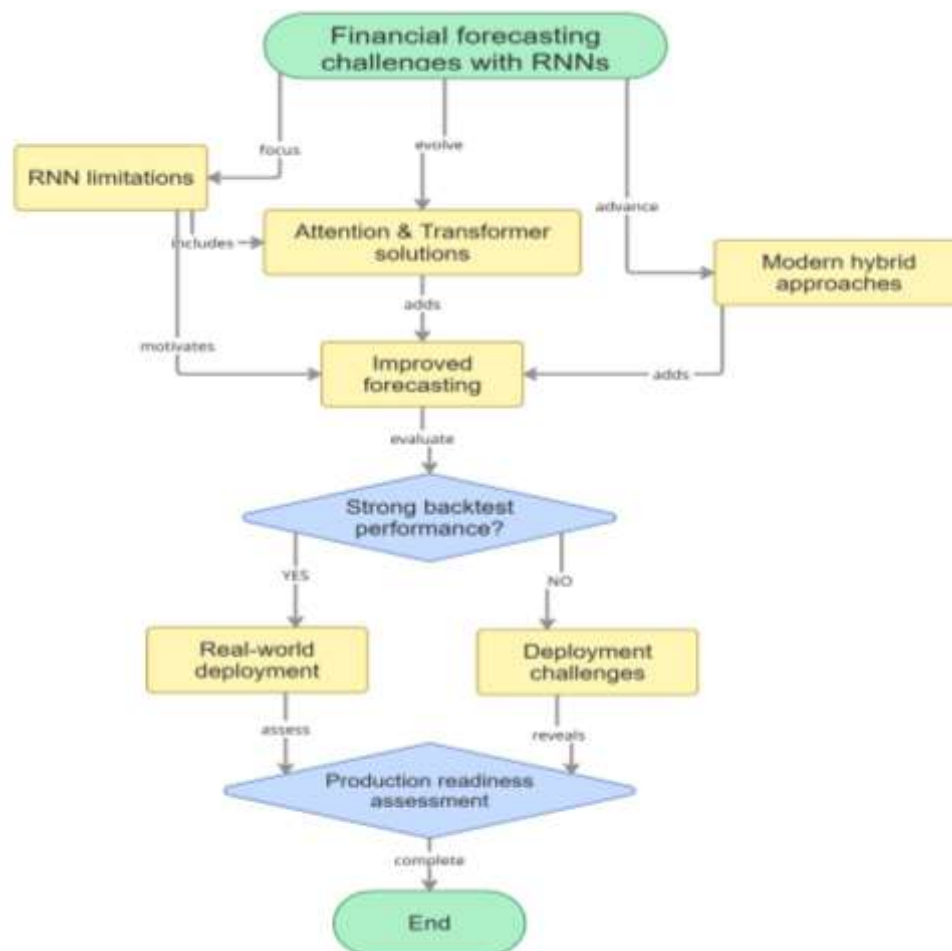


Figure 3 Transformer Hybrid Model Challenges

LLMs are also starting to be utilized as feature generation techniques rather than sentiment scoring methods. Sun et al. propose a novel attention based model which uses an LLM generated feature as part of the predictive model, arguing – and probably rightly so – that the internal representation of a company's context in the LLM contains information beyond a mere numeric feature set. Going one step further than just looking at sentiment on a company-by-company basis, Chen and Kawashima use a dual-transformer framework to transfer sentiment to other companies connected through networks in addition to a firm's price, which was a definite hole in previous sentiment analysis studies. They then obtain the “formulaic alphas,” which basically are just the definitions of the factors obtained through the language model, and notice how they perform better not only compared to just numbers but also to the manually crafted sentiment scores (Chen & Kawashima, Sentiment-

aware stock price prediction with transformer and LLM-generated formulaic alpha, 2026). In addition to this, in Chen et al. work, a method of embedding an LLM into an adaptive residual network for forecasting and using the LLM to directly extract enterprise relations from text and create a knowledge graph, which then will be combined with the adaptive residual network for forecasting – a design which combines sections IV-C and IV-D of this paper and might be considered more of a proof of the narrowing gap between the two fields (Chen, et al., 2026).

The issue of whether or not the LLM reasoning layer generalizes to conditions other than backtest settings is a different matter altogether. The authors Kim et al. evaluate the performance of sector-aware LLM-based financial analysts and discover a considerable difference between the performance in different sectors, something which may well be overlooked if one provides just an aggregated number, another example of the phenomenon that occurs all through the paper (Kim, Jeong, Ko, & Lee, 2026).

D. Graph-Based and Relational Prediction Systems

A separate line of work treats the market as a graph rather than a collection of independent series, on the reasoning that price movements are correlated through supply chains, sector membership, and institutional overlap. Liu, Duan, Geng, and Zhang's DishFT-GNN is representative of where this sub-field has moved: rather than modelling only historical spatiotemporal dependencies, as earlier GNN predictors did, they train a teacher–student pair in which the teacher learns the correlation between historical and future distribution shifts and distils that signal into the student network. They report accuracy gains of over 2%, up to 5.4% in the best case, and substantial improvements in Matthews correlation coefficient across S&P 100 and NASDAQ 100 data (Liu, Duan, Geng, & Zhang, 2025). Sun and Zhou take the dynamic-graph idea in a different direction, applying it specifically to overnight stock movement — a prediction window that is underserved by most daily-frequency models but arguably more decision-relevant for gap-risk management [12].

The journey being taken in this instance, from graphs that don't change with time, towards those that do, is part of a wider trend seen throughout the review, whereby recent systems are less focused on trying to predict individual stocks, but on finding structure between assets – something that directly affects an algorithm which needs to choose from a range of options.

E. Reinforcement-Learning-Based Trading and Portfolio Systems

A distinctly different family of systems neither predicts price nor trend; rather, it learns a trading or allocation strategy based on rewarding interactions with a simulated financial market environment. This family of systems, in our opinion, best fits the description of a real recommender system in that its natural output form matches that of a recommendation.

They use an unsupervised clustering algorithm, K-means, to select some candidates from the Indian market, while explicitly considering a balance between long-term returns and short-term returns. Ramya and Suresha use unsupervised clustering (K-means) for the candidate selection from the Indian market in conjunction with a reinforcement-learning weighting scheme. A few 2025–2026 papers "tweak" the reward specifically because a poorly designed reward function is widely agreed to be the most important difference between a policy that gets the same return as past strategies

and one that does a better job of managing risks. They take three different DRL agents to receive training on log returns, differential Sharpe ratio and maximum drawdown respectively, and then combine them into a single policy that learns the risk-adjusted reward (Choudhary, Orra, Sahoo, & Thakur, 2025) (an explicit admission that no single reward captures the reward that a risk-averse investor actually desires). The same reason is the reason of the inclusion of the downside risk in the reward function of Yang, Wang, and Hu's RiskawareTrader (Yang, Wang, & Hu, 2025), where the construction of the portfolio is modelled as a Markov Decision Process biased towards capital preservation for risk-averse users. Wei, Chen, Zhang, and co. approach the architecture side with a mixture-of-experts DRL portfolio architecture that introduces spatio-temporal attention to capture stock correlations, and dynamically routes to the best expert policy for the current market conditions, which is a frequent criticism of a fixed policy that can't adapt to regime changes (Wei, et al., 2025).

The persistent limitation across this entire family, and one that none of the papers above fully resolve, is the gap between simulated backtest performance and live execution, where slippage, liquidity constraints, and non-stationary reward surfaces erode advantages that looked robust offline.

Another, lesser-known constraint that should be mentioned is the reward engineering trial and error that is still the main approach in this literature, which is often couched in the language of principle. The choice of three distinct agents trained on each reward function is, optimistically, a concession that there is no single scalar reward which adequately reflects investor preference and pessimistically, a loophole for the difficulty of multi-objective RL in continuous action spaces (Choudhary, Orra, Sahoo, & Thakur, 2025). Downside-risk-weighted reward function of Yang, Wang, and Hu is a sensible solution, for a particular investor archetype (the risk averter), but not a general solution to this problem (Yang, Wang, & Hu, 2025). This is not a criticism on these two papers — it's almost a rule of thumb — but this does mean that designing a recommender system on top of this family should take the 'reward function' as a top-tier design consideration that needs to be validated as a separate design choice, rather than as a relatively uninteresting implementation matter to be settled after deciding on the architecture.

F. Explainability as a Design Requirement

A distinct and fast-growing thread treats interpretability as functional rather than diagnostic, partly regulatory pressure (asset managers cannot act on a recommendation they cannot justify to a compliance committee), and partly a genuine research question about whether black-box models learn economically sensible relationships or spurious correlations. Wang et al. embed SHAP-based Shapley decomposition directly into their forecasting pipeline and find that the resulting factor rankings for Chinese A-shares differ meaningfully from the profitability- and value-dominated hierarchy reported for U.S. equities in earlier asset-pricing work, with liquidity and short-horizon reversal signals dominating instead, a market-structure-dependent result a black-box model would not have surfaced on its own (Wang, Xie, Duan, Xiong, & Chen, 2026). This might be one of the best arguments for explainability throughout the review because it goes beyond the need to satisfy a compliance officer; it is about finding a model that will learn a factor structure that otherwise won't be questioned.

One of the main criticisms of price-only and even sentiment-fusion models is that they are still using sentiment in a static way, without any conditions applied. What the sentiment literature discussed in Section IV-C suggests is that sentiment helps more on days when there are some significant events like earnings announcements and macro announcements, but not consistently on all trading days..

G. Recommender-System Approaches Specific to Financial Assets

A few systems that explicitly state that stock selection is a recommendation problem are less developed but most closely related to the framework presented in Section III. While pure price-prediction systems never have to face a similar dilemma of which external knowledge sources truly boost the ranking of investable assets, rather than just the point forecast of any one of them, Guo, Sanz-Cruzado, and McCreadie address this question for non-personalized financial asset recommendation (Guo, Sanz-Cruzado, & McCreadie, 2025). Lee, Kim, Sanz-Cruzado, McCreadie, and Lee focus on personalization by recommending stocks that are personalized to a specific return-risk profile of a given investor: One of the very few systems that have been reviewed here which optimizes both MVE and personalization in a single step, rather than as separate optimization problems. Sambare, Deore, Itkar, Jadhav and Joshi approach the problem from a more practical side, and propose that instead of short-horizon trading signals, the recommendations are to be used for long-term investment decision making based on AI/ML technology, which is a reminder that "recommendation" in this context means very different investment horizons and risk appetites (Sambare, Deore, Itkar, Jadhav, & Joshi, 2026).

This helps reframe the design challenge in a productive manner. A forecasting model is designed to be optimal for reducing error on a continuous or categorical outcome, whereas a recommender is optimal for ranking based on quality, diversification, and ultimately personalized to the constraints of an investor, which the accuracy-focused research in Sections IV-A to IV-D was not designed to capture. The scarcity of finance-related recommenders compared to purely predictive models highlights another one of the key gaps identified by this review.

V. Comparative Analysis and Discussion

Table I summarizes where each family is positioned on the two axes introduced in Section III. Predictive sophistication is scored qualitatively as low, medium, or high based on the reviewed evidence; recommendation-readiness reflects whether the family's typical output already resembles a ranked, portfolio-aware action set.

TABLE I. Comparison of Methodological Families

Family	Predictive Sophistication	Recommendation-Readiness	Representative Work
Classical ML / statistical	Low–Medium	Low	(Guo et al., 2025)
Deep sequence models (LSTM/GRU)	Medium	Low	(Kiruthika, 2025), (Chen & Kawashima, 2026)

Attention / transformer / LLM-augmented	High	Low–Medium	(Wang et al., 2026), (Kayit & Ismail, 2025), (Nejatbakhsh & Aliasgari, 2025), (Sun et al., 2026), (Chen & Kawashima, 2025), (Chen et al., 2026)
Graph-based / relational	High	Medium	(Liu et al., 2025), (Kim et al., 2026)
Reinforcement-learning trading	Medium–High	High	(Ramya & Suresha, 2024), (Sun & Zhou, 2025), (Choudhary et al., 2025), (Yang et al., 2025)
Explainable AI layer	N/A (cross-cutting)	Medium (enables trust)	(Kiruthika, 2025)
Recommender-specific	Medium	High	(Lee et al., 2024), (Wei et al., 2025), (Sambare et al., 2026)

Thirdly, explainability of a model has evolved from being a desirable feature to a must-have feature due to regulatory constraints and necessity to eliminate the models attempting to fit the noise, but not the signal (Wang, Xie, Duan, Xiong, & Chen, 2026). Nevertheless, in its current implementation in the majority of reviewed papers, explainability is an additional attribute attached to a prediction model, but not a constraint determining the choice of a prediction model right from the start. Fourthly, and in direct relevance to the framework proposed in Section III, the approaches that tackle the problem as recommendation are still less popular compared to prediction-only approaches (Lee Y. , Kim, Sanz-Cruzado, McCreadie, & Lee, 2024; Guo, Sanz-Cruzado, & McCreadie, 2025; Sambare, Deore, Itkar, Jadhav, & Joshi, 2026). Architectural improvements reviewed in Sections IV-B through IV-D have not yet been translated to recommendation-focused design with the corresponding handling of the quality of recommendations, portfolio diversification, and personalization to the risk tolerance of an investor. That is exactly the gap – a mature prediction layer on top of a less advanced recommendation layer – that justifies the focus on recommendation as a primary design goal instead of an auxiliary aspect resulting from the choice of a prediction model of the year.

The fifth pattern becomes clear when the families are juxtaposed, not read independently, and is not fully understood until Table I is sorted: the families with the highest scores on recommendation-readiness (reinforcement learning, recommender-specific systems) tend to be scored on more and less consistent scales than the families with the highest scores on predictive sophistication. The attention-based and graph-based predictors are consistently evaluated in the context of the S&P 500 or NASDAQ 100 or similar large indices; the recommender-specific systems in Section IV-G are evaluated on proprietary datasets which are more difficult to compare across papers, e.g. FAR-Trans-style investor transaction logs, a specific national market, a snapshot of a knowledge-graph. This is not necessarily a defect in the individual papers (several are even explicit in their realization that they have a limitation here), but it does mean that statements of the quality of recommendation in this literature are currently based on a less solid empirical foundation than statements of predictive

accuracy, and that the two types of claims should be interpreted accordingly, rather than interchangeably.

VI. Open Challenges and Future Research Directions

A number of open questions are raised in the course of the analysis above.

Unifying forecasting and portfolio recommendation. Hardly any of the reviewed frameworks optimize prediction accuracy and portfolio-level criteria simultaneously, like diversification and sector constraints. Reinforcement learning approaches have an innate advantage, yet even within them the reward function used (Section IV-E) is typically a single-objective approximation of a truly multi-objective investor's preference function.

Event-dependent sentiment weighting. The evidence from Sections IV-C and IV-F indicates sentiment predictors are particularly valuable at moments when discrete events take place. However, the vast majority of frameworks incorporate sentiment information as a constant signal rather than applying some event gating.

Interpretable models that constrain the choice of a model rather than interpretation of outputs. Post hoc SHAP or LIME explanations reveal the workings of a model trained with specific data but do nothing to stop the designer from employing an interpretable model capable of recognizing a pattern in noise. Interpretability constraints should be incorporated into model selection stage but haven't been studied in much detail so far.

Gap between backtest and live execution. The most common limitation noted throughout the reinforcement-learning family literature, and one that should get much more coverage across all the families, is that backtesting processes cannot fully account for slippage, liquidity issues, or non-stationary regimes in the markets regardless of the sophistication of the model behind them.

Personalization at scale. Some work has been done within the recommender family (Section IV-G) to personalize recommendation systems according to the individual user's risk tolerance, but this type of work remains relatively sparse, with no system attempting such personalization based on both risk tolerance, holding period, and portfolio composition.

LLM reliability and evaluation. Systems using language learning models (LLMs) (Section IV-C) have potential, but they exhibit some variability depending on the sector (Kim, Jeong, Ko, & Lee, 2026), and there is also a lack of standardized evaluation methods for LLM-derived features and reasoning chains in finance.

VII. Conclusion

As can be seen from the review of the literature above, the story that emerges in terms of financial forecasting research is quite coherent: financial forecasting has been getting ever more architecturally complex, ever more relational, ever more sentiment-aware, and ever more LLM-aware and, rather late in the day, ever more concerned with explainability and economic realism. The one thing that financial forecasting has not done yet, and consistently, is focus on recommending. The decision an investor needs to make is really about ranking and sizing.

The framework of analysis presented in Section III is intended as an approach to making the gap explicit rather than to resolving it directly. A chart plotting the eight methodological frameworks discussed in this paper along the axes of predictive sophistication and recommendation readiness makes evident that the two are not positively correlated in a way that might be expected – while the most sophisticated from a predictive standpoint (attention-enhanced and LLM-assisted architectures) are not the most recommendation-ready, reinforcement-learning and graph-based systems, being less complex architecturally in some respects, appear to more resemble a recommendation's actual form. The recommendation engine worth developing will require taking advantage of the predictive power described in Sections IV-B to IV-D, the risk-aware reward system of Section IV-E, and the explanatory power of Section IV-F, while considering ranking performance, diversification, and personalization as its main criteria of success.

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